

MONTE SANT'ANGELO, Italy

WMO: 162580

Lat: 41.708N

Lon: 15.948E

Elev: 848

StdP: 91.54

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.9	(c) -2.2	(d) -15.1	(e) 1.1	(f) 6.8	(g) -11.4	(h) 1.6	(i) 4.4	(j) 16.6	(k) -1.2	(l) 14.2	(m) 1.2	(n) 7.2	(o) 340	(p) 0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.9	(c) 30.6	(d) 17.2	(e) 29.0	(f) 16.9	(g) 27.6	(h) 16.6	(i) 20.3	(j) 25.3	(k) 19.4	(l) 24.7	(m) 18.6	(n) 24.2	(o) 2.8	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	15.2	22.2	17.7	14.1	21.6	16.7	13.2	21.0	62.5	25.3	59.2	24.7	56.4	24.2	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.7	10.8	9.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-5.9	32.5	1.7	2.2	-7.2	34.1	-8.1	35.4	-9.1	36.6	-10.3	38.2
			-7.0	23.2	1.4	1.4	-8.0	24.2	-8.8	25.0	-9.6	25.8	-10.6	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	4.0	4.4	7.0	10.5	14.6	19.3	22.1	22.2	17.3	13.4	9.4	5.4
	DBStd	7.41	3.30	3.55	3.85	3.54	4.17	4.38	3.71	3.62	3.74	3.67	3.33	3.37
	HDD10.0	706	187	161	109	33	6	1	0	0	1	13	49	147
	HDD18.3	2499	443	391	353	234	132	39	9	8	63	156	269	402
	CDD10.0	1619	1	3	15	49	148	281	374	379	219	118	30	3
	CDD18.3	374	0	0	0	1	15	70	124	129	32	3	0	0
	CDH23.3	2108	0	0	0	0	44	362	779	804	117	1	0	0
	CDH26.7	512	0	0	0	0	6	81	194	213	18	0	0	0
(14) Wind	WSAvg	4.2	4.8	5.2	4.8	4.3	3.9	3.6	3.5	3.5	3.7	3.8	4.2	4.9
(15) Precipitation	PrecAvg	693	77	50	57	57	47	49	50	38	72	66	65	77
	PrecMax	1570	333	181	147	204	170	150	187	95	215	182	155	221
	PrecMin	288	1	8	8	8	1	1	0	0	1	1	7	15
	PrecStd	302	78	50	45	45	39	43	47	28	57	52	42	57
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	14.6	17.0	21.0	27.4	31.4	32.8	32.6	28.9	22.7	18.6	13.5
		MCWB	8.1	7.0	10.1	12.2	15.7	18.2	17.1	16.9	16.7	15.9	10.9	8.7
	2%	DB	11.0	12.3	15.1	19.0	24.3	29.0	30.4	30.8	26.4	20.9	16.1	11.9
		MCWB	7.1	6.4	9.0	10.8	14.6	17.2	17.3	17.0	16.4	14.1	11.1	8.0
	5%	DB	9.6	10.3	13.7	17.3	22.4	27.1	28.8	29.1	24.4	19.4	14.7	10.8
		MCWB	6.5	6.3	8.2	10.0	14.1	17.0	16.8	16.9	15.9	13.9	10.9	7.5
	10%	DB	8.3	8.9	12.3	15.6	20.5	25.4	27.2	27.4	22.4	18.1	13.3	9.7
		MCWB	5.9	6.1	7.5	9.3	13.2	16.0	16.3	16.6	15.6	13.2	10.5	7.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.3	11.9	13.4	17.6	21.0	21.4	21.8	20.2	17.5	14.5	10.5
		MCDB	12.4	11.4	14.8	19.0	23.5	26.1	26.0	25.2	23.7	19.9	15.8	12.2
	2%	WB	8.5	8.2	10.4	12.2	16.2	19.4	20.0	20.4	18.7	16.2	13.0	9.4
		MCDB	9.8	10.1	13.3	16.8	21.6	26.0	25.5	24.7	22.1	18.7	14.4	10.7
	5%	WB	7.4	7.3	9.3	11.2	15.1	18.3	19.2	19.4	17.8	15.3	12.0	8.5
		MCDB	8.7	9.2	12.1	15.3	20.8	24.6	25.0	24.5	21.9	17.9	13.4	10.0
	10%	WB	6.3	6.4	8.4	10.2	14.1	17.3	18.3	18.6	16.7	14.4	11.1	7.6
		MCDB	7.7	8.3	11.2	13.9	19.4	23.5	24.3	24.1	20.6	16.8	12.5	9.2
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	4.2	4.9	5.7	6.2	6.5	6.7	6.9	5.6	4.8	4.0	3.7
		MCDBR	5.5	6.1	6.3	6.7	7.6	7.9	7.7	7.7	7.0	5.9	5.5	4.7
		MCWBR	4.8	4.7	5.1	5.6	5.4	5.8	5.2	5.1	5.8	5.5	4.9	4.2
	5% WB	MCDBR	4.7	5.1	5.6	6.3	7.2	7.3	7.1	7.2	6.2	5.3	4.6	3.9
		MCWBR	4.6	4.5	4.9	5.6	5.6	6.1	5.9	6.1	6.0	5.5	4.6	4.1
(40) Clear-Sky Solar Irradiance	taub		0.307	0.327	0.354	0.398	0.395	0.390	0.387	0.389	0.377	0.350	0.329	0.308
	taud		2.514	2.444	2.372	2.245	2.313	2.355	2.367	2.364	2.398	2.469	2.482	2.516
	Ebn at Noon		849	880	891	871	881	883	882	871	857	842	810	815
	Edh at Noon		72	90	109	133	127	122	120	116	105	87	74	67
(44) All-Sky Solar Radiation	RadAvg		1.72	2.49	3.81	5.08	6.24	6.99	7.13	6.27	4.53	3.14	1.98	1.53
	RadStd		0.20	0.25	0.37	0.40	0.42	0.34	0.34	0.38	0.26	0.27	0.16	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-100	+144	+67

Nomenclature: See separate page